

Andrew Freeman

Montreal, QC · 514-972-5682 · andrewfreeman803@gmail.com

Seeking Winter 2027 – Summer 2027 internship

Canadian Citizen – Open to work in U.S. (requires visa sponsorship)

[LinkedIn](#)

[Project Portfolio](#)

Mechanical Engineering student (Concordia, 2028) with 3+ years designing and validating high-voltage battery systems, thermal models, and electromechanical hardware integration for competitive Formula SAE electric vehicles. Built a 369.6V pack from cell selection through ANSYS-validated thermal models, then raced it. Fluent in Python, MATLAB, SolidWorks, and hands-on lab instrumentation.

TECHNICAL SKILLS

Hardware & Test: Electronic loads • Oscilloscopes • DCIR/HPPC/capacity testing • Thermal instrumentation • Tensile test fixtures (ASTM)

Simulation: ANSYS Fluent (CHT, airflow) • RC electrical models • MATLAB (pump curves, thermal, electrical)

CAD: SolidWorks • Fusion 360 • AutoCAD (DFM)

Programming: Python (automation, data logging, control scripts) • MATLAB (Battery models) • C, C++ (embedded systems)

Systems: Full-vehicle electromechanical integration (HV battery, BMS, cooling, chassis, suspension, drivetrain, LV electrical, vehicle controls) • Sensor integration (analog/digital) • GDB firmware debugging

OS: Windows • Linux

Languages: English — fluent • French — fluent • Spanish — fluent

EDUCATION

B.Eng. Mechanical Engineering · *Concordia University (GPA 3.75/4.3)*

Dean's Honour List

2022 – Present

Expected Graduation May 2028

Relevant courses: Heat Transfer · Machine Element Design · Microcontrollers · Properties & Failures of Material · Control Systems

EXPERIENCE

Technical Director/High Voltage Lead · *Concordia Formula Racing*

Jun 2024 – Present

Led all engineering sub-teams (chassis, drivetrain, suspension, HV/LV electrical) approximately 20 dedicated members (40 general members) and a budget of roughly \$40 000 CAD for an electric FSAE vehicle drove decisions in managing personnel and budgets across all mechanical and electrical subsystems.

- ▶ Designed a 369.6 V lithium-ion battery pack end-to-end: cell selection, enclosure, BMS integration and validation (replacing a prior design)
- ▶ Cut cell-characterization timeline by 40% by authoring Python scripts that automated DCIR/capacity test cycles on electronic loads, eliminating manual data transcription.
- ▶ Built a coupled RC-electrical + conjugate heat-transfer thermal model (MATLAB + ANSYS Fluent) that predicted pack temperature within ± 3 °C across endurance-simulation runs, reducing physical test iterations.
- ▶ Designed a lap-level regen brake-blending model (MATLAB, MoTeC telemetry) and implemented the real-time torque arbitration controller in embedded C on the rear vehicle controller blending regen with mechanical braking under six layered limits (SOC, charge-current, motor-torque, ABS-lockup)
- ▶ Validated structural enclosure materials at operating temperatures: designed furnace-heated tensile test fixtures per ASTM standards, confirming yield-strength and Young's modulus assumptions used in CAD safety margins.

- Identified and corrected a critical electrical fault in vehicle control boards via GDB-assisted firmware debugging, preventing DNF at competition.
- Led design reviews across chassis, suspension, and drivetrain enforcing FOS targets and re-running FEA with real boundary conditions (e.g., bolt preload) to catch discrepancies before manufacturing and curb over-engineering.
- Redesigned and personally welded the chassis to house a battery pack 4" smaller in both length and width, reducing wheelbase from 1,575 mm to 1,540 mm (tighter turning radius) and cutting chassis mass by 1.3 kg while satisfying all FSAE structural and packaging regulations.
- Developed and validated a 2 RC model for SOC estimation. Tracking within 1% SOC throughout range.

Cooling System Lead · *Concordia Formula Racing*

Jun 2023 – Jun 2024

Owned thermal design for the team's first competitive EV, with zero prior baseline to work from.

- Delivered a cooling system over 50% lighter than the previous combustion-era design by building a MATLAB pump-selection model that optimized head curves, pressure losses, and duty cycle simultaneously.
- Designed and integrated a real-time DAQ system using thermistors and Hall-effect flow sensors, enabling instantaneous kW heat-rejection measurement - data that directly informed the 2024 pack thermal model.
- Optimized pump duty cycle through the MATLAB model, reducing average cooling-system current draw and extending projected battery range under endurance-event conditions.

Pit Crew Technician · *SH-Karting*

Jul 2024 – Sep 2025

- Maintained mechanical reliability of Porsche 944 and 911 race vehicles under continuous hot lap conditions; rapid trackside diagnostics including brake service and EFI/carb fuel-mixture correction, keeping vehicle downtime near zero.

Lab Intern · *Nitrex Metal Inc.*

May 2023 – Jan 2025

- Led a salt-bath corrosion-testing project evaluating seven heat-treatment process variants; statistical analysis of mechanical-property variance provided actionable process-optimization recommendations adopted by the QA team.
- Maintained traceability documentation for a commercial heat-treatment facility operating under strict quality-compliance protocols, achieving zero audit findings across tenure.

PROJECTS

Hardware-in-the-Loop vehicle controller · *MECH 471 Microcontrollers*

2025

- Built a real-time RC-car controller on an Arduino Uno (ATmega328P) running a 50 Hz deterministic control loop, closing the loop against a separate HIL plant board (vehicle + tire model) over GPIO: three analog wheel speed inputs in, two servo-style command pulses out.
- Wrote a custom register level hardware abstraction layer in C instead of using the Arduino library. Made timer driven servo PWM via hardware output compare interrupts, a buffered multi-channel ADC driver (/128 prescaler, 125 kHz), digital I/O, and a reusable PID library.

Structured the project as a multi-target PlatformIO build tying the controller and simulator firmware together; full source on GitHub.

Amazon DeepRacer — Reinforcement Learning · *Vanier College*

2022

- Trained an RL driving model on a self-hosted Linux server, combining vehicle-dynamics constraints with reward-function tuning; finished 67th out of 2,325 entrants (top 3%) in the unranked division.
- Made custom reward functions to force model on track and obtain faster and faster lap times.